ANNUALIZED RETURN

Learning Outcome

- calculate and interpret annualized return measures and continuously compounded returns, and describe their appropriate uses

The period during which a return is earned or computed can vary and often we have to annualize a return that was calculated for a period that is shorter (or longer) than one year. You might buy a short-term treasury bill with a maturity of three months, or you might take a position in a futures contract that expires at the end of the next quarter. How can we compare these returns?

In many cases, it is most convenient to annualize all available returns to facilitate comparison. Thus, daily, weekly, monthly, and quarterly returns are converted to annualized returns. Many formulas used for calculating certain values or prices also require all returns and periods to be expressed as annualized rates of return. For example, the most common version of the Black–Scholes option-pricing model requires annualized returns and periods to be in years.

Non-annual Compounding

Recall that interest may be paid semiannually, quarterly, monthly, or even daily. To handle interest payments made more than once a year, we can modify the present value formula as follows. Here, R_s is the quoted interest rate and equals the periodic interest rate multiplied by the number of compounding periods in each year. In general, with more than one compounding period in a year, we can express the formula for present value as follows:

$$\mathrm{PV} = \mathrm{FV}_N \left(1 + \frac{R_s}{m}\right)^{-mN}, \tag{7}$$

where

m = number of compounding periods per year,

R_s = quoted annual interest rate, and

N = number of years.

The formula in Equation 7 is quite similar to the simple present value formula. As we have already noted, present value and future value factors are reciprocals. Changing the frequency of

compounding periods.

The following example presents an application of monthly compounding.

EXAMPLE 11

The Present Value of a Lump Sum with Monthly Compounding

The manager of a Canadian pension fund knows that the fund must make a lump-sum payment of CAD5 million 10 years from today. She wants to invest an amount today in a guaranteed investment contract (GIC) so that it will grow to the required amount. The current interest rate on GICs is 6 percent a year, compounded monthly.

1. How much should she invest today in the GIC?

Hide Solution

Solution:

By applying Equation 7, the required present value is calculated as follow:

$$\text{FV}_N = \text{CAD } 5{,}000{,}000$$

$$R_s = 6\% = 0.06$$

$$m = 12$$

$$R_s/m = 0.06/12 = 0.005$$

$$N = 10$$

$$mN = 12(10) = 120$$

$$\text{PV} = \text{FV}_N\left(1 + \frac{R_s}{m}\right)^{-mN}$$

$$= CAD\ 5{,}000{,}000(1.005)^{-120}$$

$$= CAD\ 5{,}000{,}000(0.549633)$$

$$= CAD\ 2{,}748{,}163.67$$

In applying Equation 7, we use the periodic rate (in this case, the monthly rate) and the appropriate number of periods with monthly compounding (in this case, 10 years of monthly compounding, or 120 months).

Annualizing Returns

periods in a year. A monthly return is compounded 12 times, a weekly return is compounded 52 times, and a quarterly return is compounded 4 times. Daily returns are normally compounded 365 times. For an uncommon number of days, we compound by the ratio of 365 to the number of days.

If the weekly return is 0.2 percent, then the compound annual return is 10.95 percent (there are 52 weeks in a year):

$$R_{annual} = \left(1 + R_{weekly}\right)^{52} - 1 = (1 + 0.2\%)^{52} - 1$$
$$= (1.002)^{52} - 1 = 0.1095 = 10.95\%$$

If the return for 15 days is 0.4 percent, then the annualized return is 10.20 percent, assuming 365 days in a year:

$$R_{annual} = (1 + R_{15})^{365/15} - 1 = (1 + 0.4\%)^{365/15} - 1$$
$$= (1.004)^{365/15} - 1 = 0.1020 = 10.20\%$$

A general equation to annualize returns is given, where *c* is the number of periods in a year. For a quarter, *c* = 4 and for a month, *c* = 12:

$$R_{annual} = \left(1 + R_{period}\right)^{c} - 1. \quad 8$$

How can we annualize a return when the holding period return is more than one year? For example, how do we annualize an 18-month holding period return? Because one year contains two-thirds of 18-month periods, *c* = 2/3 in the above equation. For example, an 18-month return of 20 percent can be annualized as follows:

$$R_{annual} = (1 + R_{18month})^{2/3} - 1 = (1 + 0.20)^{2/3} - 1 = 0.1292 = 12.9$$

Similar expressions can be constructed when quarterly or weekly returns are needed for comparison instead of annual returns. In such cases, *c* is equal to the number of holding periods in a quarter or in a week. For example, assume that you want to convert daily returns to weekly returns or annual returns to weekly returns for comparison between weekly returns. To convert daily returns to weekly returns, *c* = 5, assume that there are five trading days in a week. However, daily return calculations can be annualized differently. For example, five can be used for trading-day-based calculations, giving approximately 250 trading days a year; seven can be used on calendar-day-based calculations. Specific methods used conform to specific business practices, market conventions, and standards. To convert annual returns to weekly returns, *c* = 1/52. The expressions for annual returns can then be rewritten as expressions for weekly returns as follows:

$$R_{weekly} = \left(1 + R_{daily}\right)^{5} - 1; R_{weekly} = (1 + R_{annual})^{1/52} - 1. \quad 9$$

can be reinvested repeatedly while earning a similar return. This type of return is not always possible. An investor may earn a return of 5 percent during a week because the market rose sharply that week, but it is highly unlikely that he will earn a return of 5 percent every week for the next 51 weeks, resulting in an annualized return of 1,164.3 percent (= $1.05^{52} - 1$). Therefore, it is important to annualize short-term returns with this limitation in mind.

EXAMPLE 12

Annualized Returns

An analyst seeks to evaluate three securities she has held in her portfolio for different periods of time.

- Over the past 100 days, Security A has earned a return of 6.2 percent.
- Security B has earned 2 percent over the past four weeks.
- Security C has earned a return of 5 percent over the past three months.

1. Compare the relative performance of the three securities.

Hide Solution

Solution:

To facilitate comparison, the three securities' returns need to be annualized:

- Annualized return for Security A: $R_{SA} = (1 + 0.062)^{365/100} - 1 = 0.2455 = 24.55\%$
- Annualized return for Security B: $R_{SB} = (1 + 0.02)^{52/4} - 1 = 0.2936 = 29.36\%$
- Annualized return for Security C: $R_{SC} = (1 + 0.05)^{4} - 1 = 0.2155 = 21.55\%$

Security B generated the highest annualized return.

EXAMPLE 13

Exchange-Traded Fund Performance

An investor is evaluating the returns of three recently formed exchange-traded funds. Selected return information

Exhibit 16: ETF Performance Information

ETF	Time Since Inception	Return Since Inception (%)
1	146 days	4.61
2	5 weeks	1.10
3	15 months	14.35

1. Which ETF has the highest annualized rate of return?

A. ETF 1
B. ETF 2
C. ETF 3

Hide Solution

Solution:

B is correct. The annualized rate of return for the three ETFs are as follows:

ETF 1 annualized return = $(1.0461^{365/146}) - 1 =$ 11.93%

ETF 2 annualized return = $(1.0110^{52/5}) - 1 =$ 12.05%

ETF 3 annualized return = $(1.1435^{12/15}) - 1 =$ 11.32%

Despite having the lowest value for the periodic rate, ETF 2 has the highest annualized rate of return because of the reinvestment rate assumption and the compounding of the periodic rate.

Continuously Compounded Returns

An important concept is the continuously compounded return associated with a holding period return, such as R_1. The **continuously compounded return** associated with a holding period return is the natural logarithm of one plus that holding period return, or equivalently, the natural logarithm of the ending price over the beginning price (the price relative). Note that here we are using r to refer specifically to continuously compounded returns, but other textbooks and sources may use a different notation.

If we observe a one-week holding period return of 0.04, the equivalent continuously compounded return, called the one-week continuously compounded return, is ln(1.04) = 0.039221; EUR1.00 invested for one week at 0.039221 continuously

The continuously compounded return from t to $t + 1$ is

$$r_{t,t+1} = \ln(P_{t+1}/P_t) = \ln(1 + R_{t,t+1}). \quad (10)$$

For our example, an asset purchased at time t for a P_0 of USD30 and the same asset one period later, $t + 1$, has a value of P_1 of USD34.50 has a continuously compounded return given by $r_{0,1} = \ln(P_1/P_0) = \ln(1 + R_{0,1}) = \ln(\text{USD}34.50/\text{USD}30) = \ln(1.15) = 0.139762$.

Thus, 13.98 percent is the continuously compounded return from $t = 0$ to $t = 1$. The continuously compounded return is smaller than the associated holding period return. If our investment horizon extends from $t = 0$ to $t = T$, then the continuously compounded return to T is

$$r_{0,T} = \ln(P_T/P_0). \quad (11)$$

Applying the exponential function to both sides of the equation, we have $\exp(r_{0,T}) = \exp[\ln(P_T/P_0)] = P_T/P_0$, so

$$P_T = P_0\exp(r_{0,T}).$$

We can also express P_T/P_0 as the product of price relatives:

$$P_T/P_0 = (P_T/P_{T-1})(P_{T-1}/P_{T-2}) \ldots (P_1/P_0). \quad (12)$$

Taking logs of both sides of this equation, we find that the continuously compounded return to time T is the sum of the one-period continuously compounded returns:

$$r_{0,T} = r_{T-1,T} + r_{T-2,T-1} + \ldots + r_{0,1}. \quad (13)$$

Using holding period returns to find the ending value of a USD1 investment involves the multiplication of quantities (1 + holding period return). Using continuously compounded returns involves addition (as shown in Equation 13), which is a desirable property of continuously compounded returns and which we will use throughout the curriculum.

Discuss Filter

Discussion

What do you want to discuss... (required)

TV

Hi guys might have a simple question but I don't seem to understand ---- In the formula for annualizing returns, why is the ratio of days (e.g., 365/146) placed inside the exponent, like this:
(1R 365/146)−1 instead of applying the division after exponentiation, like:
(1+R)365/146−1 like in the course -- I understand the first one calculates the number of periods in a year and compounds accordingly, but I want to be sure why the division must be inside the exponent and not outside. Could you please explain the intuition behind this?

FA

For example 13 I am getting the 2nd ETF = 10.902, how it mention 12.05 ?

Created a month ago by Fatma Aghaei 2 replies | Last Activity: a month ago

Show All replies Reply to this Comment

JB

hi guys, I'm a little confused — the practical tests for most topics seem quite closely tied to the content we've learned, but about half of the ones for this topic feel quite different. Were those parts covered elsewhere, or are we meant to draw on prior knowledge? Just want to make sure I haven't overlooked anything. Thanks in advance! 🙏

Created 3 months ago by Jianan Bi 4 replies | Last Activity: 3 months ago

Show All replies Reply to this Comment

Example 12. Why is security C raised to the power 4?

Created 6 months ago by AHMAD SAID NJI TETGNU MBOUOMBOUO 3 replies | Last Activity: 4 months ago

Show All replies Reply to this Comment

PM

great work

Created 4 months ago by Prince Mel Gautier Mboungou Batantou 0 replies | Last Activity: 4 months ago

Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.